AofA 2026 – Program and Abstracts

22-26 June 2026

LMU Munich, Institute of Mathematics, Theresienstr. 39, D-80333 Munich

	Monday	Tuesday	Wednesday	Thursday	Friday
9:00-10:00	*Invited Talk* Bruno Salvy	*Invited Talk* Will Perkins	*Flajolet Lecture* Hsien-Kuei Hwang	*Invited Talk* Petra Berenbrink	*Invited Talk* Matthew Kwan
10:00-10:30	Coffee Break	Coffee Break	Coffee Break	Coffee Break	Coffee Break
10:30-12:30	Session A	Session C	Session E	Session F	Session H
12:30-13:30	Lunch	Lunch	Lunch	Lunch	
13:30-14:30	*Invited Talk* Eric Fusy	*Invited Talk* Irene Marcovici	Excursion	*Invited Talk* Michal Opler	
14:30-15:00	Coffee Break	Coffee Break		Coffee Break	
15:00-17:00	Session B	Session D Break		Session G	
17:00-19:00		Business Meeting			

Monday 22 June 2026, Room B006 (Institute of Mathematics)

9:00-10:00	**Invited Speaker** *Bruno Salvy* TBA
10:00-10:30	Coffee Break
	Session A
10:30-11:00	*Jacob Lundblad and Stephan Wagner* Path length and external path length in random trees
11:00-11:30	*Nadja Azzouz, Olivier Bodini, Francis Durand and Bernhard Gittenberger* Efficient sampling of increasing trees
11:30-12:00	*Philipp Beltran, Benedikt Stufler, Louigi Addario-Berry and Paul Thévenin* Scaling limits of multitype Bienaymé trees
12:00-12:30	*Michael Fuchs and Tsan-Cheng Yu* Semi-Simplex Phylogenetic Networks: Tree-Child Networks and Galled Trees
12:30-13:30	Lunch
13:30-14:30	**Invited Speaker** *Eric Fusy* TBA
14:30-15:00	Coffee Break
	Session B
15:00-15:30	*Michael Drmota and Yitian Wang* Local Central Limit Theorems for Subgraph Counts in Subcritial Graph Families
15:30-16:00	*Fabian Burghart* Ancestries and descendants in a random DAG
16:00-16:30	*Nicolas Tokka* Local Limit of Random Regular Bipartite Planar Maps
16:30-17:00	*Jeremie Bettinelli and Dimitri Korkotashvili* Link between bipartite and general unicellular toroidal maps via slit–slide–sew bijections

Tuesday 23 June 2026, Room B006 (Institute of Mathematics)

9:00-10:00	**Invited Speaker** *Will Perkins* Sampling algorithms, asymptotic enumeration, and statistical physics
10:00-10:30	Coffee Break
	Session C
10:30-11:00	*Niccolò Bosio, Benedikt Stufler and Markus Kuba* Gibbs partitions and lattice paths
11:00-11:30	*Benedikt Stufler* Poisson-Dirichlet graphons and permutons
11:30-12:00	*Michael Drmota and Zéphyr Salvy* Asymptotic Transfer in Critical Recursive Composition Schemes
12:00-12:30	*Hexuan Liu, Michael Wallner and Guan-Ru Yu* A Combinatorial Framework for the Pons-Batle Identity: Young Tableaux, Lattice Paths, and Limit Laws
12:30-13:30	Lunch
13:30-14:30	**Invited Speaker** *Irene Marcovici* Frequency of letters in some self-descriptive sequences
14:30-15:00	Coffee Break
	Session D
15:30-15:30	*Valentin Blomer and Kai-Uwe Bux* The cost of cyclic permutations and remainder sums in the Euclidean algorithm
15:30-16:00	*Aurélien Guerder* Cycle structure of random standardized permutations
16:00-16:30	*Calum Buchanan, Fabian Burghart, Stephan Wagner and Mei Yin* On cycles in multiset permutations, parking functions, and related structures
16:30-17:00	Break
17:00-19:00	Business Meeting

Wednesday 24 June 2026
Senate room E110 (Main Building, Geschwister Scholl Pl. 1)

9:00-10:00	**Flajolet Lecture** *Hsien-Kuei Hwang* TBA
10:00-10:30	Coffee Break
	Session E
10:30-11:00	*Michael Drmota and Eva-Maria Hainzl* Singularly perturbed discrete differential equations and pattern counts in simple triangulations
11:00-11:30	*Victor Dubach, Thomas Budzinski, Valentin Féray, Mohamed Slim Kammoun and Mylene Maïda* Large deviation principles for pattern-avoiding permutations, and limit shapes for constrained Mallows permutations
11:30-12:00	*Cecilia Holmgren, Jasper Ischebeck and Svante Janson* Fringe subtrees of split trees
12:00-12:30	*Eva-Maria Hainzl* Formulas and asymptotics of hypergraph Catalan numbers
12:30-13:30	Lunch
13:30-	Excursion

Thursday 25 June 2026, Room B006 (Institute of Mathematics)

9:00-10:00	**Invited Speaker** *Petra Berenbrink* TBA
10:00-10:30	Coffee Break
	Session F
10:30-11:00	*James Allen Fill* A New Fine-scale Berry-Esseen-type Gumbel-limit Theorem for Multivariate Maxima
11:00-11:30	*Jasper Ischebeck, Florian Lesny and Ralph Neininger* A distributional analysis of QuickXsort for Mergesort
11:30-12:00	*Markus Nebel* Moment Statistics in the Boltzmann Probability Model
12:00-12:30	*Thomas Fischer and Yury Person* Fractional vs Expectation Thresholds: Random Support Case
12:30-13:30	Lunch
13:30-14:30	**Invited Speaker** *Michal Opler* From Marcus–Tardos to optimal algorithms
14:30-15:00	Coffee Break
	Session G
15:00-15:30	*Julius Hallmann, Kostas Lakis and Tamás Makai* The Dispersion Process Has the Same Phase Transition on Almost Every Graph
15:30-16:00	*Sam Olesker-Taylor, Thomas Sauerwald and Luca Zanetti* Graphical Balanced Allocations with Removals
16:00-16:30	*Geoffrey Deperle, Christine Fricker, Philippe Jacquet, Bernard Mans and Alessia Rigonat* Asymptotics of Parking Search in Hyperfractal Networks
16:30-17:00	*Ahmed Alharbi, Cyril Banderier and Charles Bouillaguet* Bounded linear probing hashing

Friday 26 June 2026, Room B006 (Institute of Mathematics)

9:00-10:00	**Invited Speaker** *Matthew Kwan* Very sparse random discrete matrices
10:00-10:30	Coffee Break
	Session H
10:30-11:00	*Guan-Huei Duh, Philipp Sprüssel and Stephan Wagner* Enumeration of bipartite acyclic digraphs
11:00-11:30	*Nadja Azzouz, Olivier Bodini, Francis Durand and Bernhard Gittenberger* Asymptotic analysis of generating functions arising from dynamic graphs
11:30-12:00	*Maximilian Wiesmann* Lee-Yang phenomena in edge-coloured graph counting

Flajolet Lecture

Title
Hsien-Kuei Hwang
Wednesday, June 24, 9:00 - 10:00

Invited Talks

Title
Bruno Salvy
Monday, June 22, 9:00 - 10:00

Title
Eric Fusy
Monday, June 22, 13:30 - 14:30

Sampling algorithms, asymptotic enumeration, and statistical physics
Will Perkins
Tuesday, June 23, 9:00 - 10:00

I will discuss connections between problems in three different fields: sampling algorithms in computer science, asymptotic enumeration in combinatorics, and phase transitions in statistical physics. The running example will be triangle-free graphs: how do you efficiently sample a uniformly triangle-free graph of a given density? How many of them are there? Is there a phase transition as the density varies? I will give an overview of what is known, what remains open, and the types of tools that are effective in tackling these problems.

Frequency of letters in some self-descriptive sequences
Irene Marcovici
Tuesday, June 23, 13:30 - 14:30

The Oldenburger-Kolakoski sequence 12211212212211211221... is the unique sequence over the alphabet 1,2 starting with 1 that is a fixed point of the run-length encoding operator. While the definition of this sequence is elementary, its combinatorial and dynamical properties remain largely unknown. In particular, Keane's question regarding the existence of the asymptotic frequency of the symbol 1, conjectured to be 1/2, is still open.

In the first part of the presentation, I will introduce probabilistic variants of the Oldenburger-Kolakoski sequence and prove results concerning the frequency of the symbol 1 in these sequences. We will then define the notion of "smooth sequence", which is another way to extend the scope of study, and discuss the properties of smooth sequences over the alphabet 1,3.

The presentation will be based on different works in collaboration with C. Boisson, D. Jamet, L. Poirier, and T. de la Rue.

Title

Petra Berenbrink

Thursday, June 25, 9:00 - 10:00

From Marcus–Tardos to optimal algorithms

Michal Opler

Thursday, June 25, 13:30 - 14:30

The Marcus–Tardos theorem, resolving the Füredi–Hajnal conjecture, has a well-known consequence: permutation classes avoiding a fixed pattern grow only singly exponentially. In information-theoretic terms, a pattern-avoiding permutation of length n carries only $O(n)$ bits of entropy, rather than the $\Theta(n \log n)$ of an arbitrary permutation. Can we exploit this decrease in complexity algorithmically?

We will describe two settings where the answer is yes, and where the algorithmic bound matches the information-theoretic optimum up to a constant factor. The first is sorting: an algorithm that sorts a length-n sequence avoiding a fixed pattern π in time proportional to its entropy, optimal up to a constant factor in both comparisons and runtime. The second is encoding: a recent joint work with László Kozma gives a compact data structure that stores such a permutation in a number of bits proportional to its entropy, while supporting rank and unrank queries in constant time. Marcus–Tardos appears in both proofs, though in quite different roles: as a bookkeeping device in the sorting algorithm, and as the source of an explicit decomposition in the data structure.

Very sparse random discrete matrices

Matthew Kwan

Friday, June 26, 9:00 - 10:00

Extremely sparse random matrices tend to be singular, due to the likely presence of "local combinatorial dependencies" such as all-zero columns or pairs of identical columns. We discuss this phenomenon, and some results showing that these kinds of combinatorial dependencies are in some sense the "only" causes of singularity. A key role is played by the leaf-removal algorithm introduced by Karp and Sipser to study the matching number of random graphs. This is joint work with Asaf Ferber, Margalit Glasgow, Ashwin Sah and Mehtaab Sawhney.

Contributed Talks